

Jacob Berger

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EDUCATION

Oregon State University — Corvallis, OR

Ph.D. & M.S. in Artificial Intelligence

Expected June 2030

Advisor: Dr. Lizhong Chen, STAR Lab. Graduate coursework: Deep Learning, Parallel Programming, Numerical ODEs/PDEs, Math of Data Science.

Honors B.S. in Applied Computer Science & Computational Mathematics

Sept 2022 – June 2026 • Summa Cum Laude (both degrees) • GPA: 3.9+

Accelerated Master's Program (graduate coursework as undergraduate). Honors degree. AI concentration, double major.

RESEARCH & PUBLICATIONS

Honors Thesis: Semantic Constraints in NLP

June 2025 – Present

Researcher (Advisor: Dr. Lizhong Chen, STAR Lab)

- First-author submission under review at AAAI 2027 ("Parsimonious Decoding"). Developed novel metrics to enforce semantic constraints in probabilistic language models, improving vocabulary legality without degrading fluency.

EXPERIENCE

AI Research Engineer

March 2023 – Present

Lexel Synergetics — Portland, OR

- Co-authored and secured a Phase I SBIR grant (federal R&D funding) for NLP-based language model development.
- Architected end-to-end data pipelines for training and fine-tuning specialized large language models (LoRA, RAG).
- Deployed cloud infrastructure on AWS, GCP, and Vercel for model training and serving.

AI Engineer & Project Manager — AuTuber

2025

Team of 4 — Multimodal AI agent for live streaming automation

- Built an LLM-driven agent that observes webcam, screen capture, and microphone audio to autonomously control OBS and VTube Studio via WebSocket APIs.
- Designed multimodal context fusion and LLM-based action planning with safety validation and cooldowns.

Head Mechanic

Dec 2022 – Present

MU Lanes and Games — Corvallis, OR

- Manage maintenance and diagnostics for electromechanical bowling lane systems.

SKILLS

Languages: Python, C++, Java, JavaScript, Bash, x86 Assembly, MATLAB

Machine Learning: PyTorch, HuggingFace Transformers, LLM Fine-tuning (LoRA), RAG Pipelines, Reinforcement Learning

Systems & Cloud: Linux (Debian), Proxmox VE, Virtualization (LXC/KVM), AWS (EC2/S3), GCP, Git, Vercel

Human Languages: Fluent Spanish (9 years immersion)

AWARDS

- Putnam Math Competition: Top 3 finisher at OSU.
- Finley Academic Excellence Scholarship: \$20,000 recipient.

EXTRACURRICULARS

- Competitive Math: OSU Putnam and COMAP Math Modeling (implemented Cellular Automata for ecosystem resilience analysis).
- Competitive Sports: Founded OSU national collegiate bowling team (All-Star qualifier); Rose City Rowing Club leader (2-time national qualifier).